

Gate 1 Write-Up — Gesture Mouse AI

Team *FutureStack* | University of Kelaniya | *IntelliCon'26*

1. Problem & Domain

Domain: AI / Computer Vision & Human-Computer Interaction (Accessibility Tech).

Traditional computer interaction depends on physical input devices such as a mouse or trackpad. This creates barriers for three overlapping groups: (1) users with limited hand mobility or physical disabilities who struggle with fine motor control; (2) people sharing devices in hygiene-sensitive environments such as hospitals, clinics, and public kiosks; and (3) presenters and educators in smart classrooms who need to control a screen from a distance. Existing touchless solutions (e.g. Leap Motion, infrared/depth-sensor kits) address this but require costly specialized hardware, putting them out of reach for the average student, educator, or small institution.

2. Target User

- Individuals with limited hand mobility who find a physical mouse difficult to operate.
- Educators and presenters who need hands-free, touch-free control of a screen during lectures or demos.
- Public kiosk and smart-classroom operators who want to avoid shared touch surfaces.

Primary early adopters: university students and staff (e.g. University of Kelaniya), assistive-technology communities, and public-sector digitalization initiatives such as Ministry of Education-funded smart classrooms.

3. Validation

Formal user testing has not yet been conducted. The problem is grounded in observed, well-documented pain points: RSI/wrist strain from prolonged mouse use, infection-control concerns around shared input devices in clinical settings, and friction for presenters tied to a workstation. Our validation plan is a structured pilot with classmates and lecturers — including at least one user with limited hand mobility — to gather feedback on gesture accuracy, comfort, and calibration ease ahead of the next iteration.

4. Market Size

The addressable market spans every laptop/desktop user seeking hands-free or hygienic input, with a near-term beachhead in Sri Lankan universities, assistive-tech communities, and public-sector digitalization programs, expanding to hospitals, museums, and public-facing kiosks. Because the product runs entirely on webcams already present in an estimated billions of laptops in use worldwide, adoption is not gated by new hardware purchases, giving the addressable base significant reach beyond niche gesture-hardware buyers.

5. Competitors

Leap Motion and similar depth-sensor / infrared gesture-control kits validate demand for touchless interaction but require dedicated external hardware, raising cost and limiting adoption to specialized labs. Existing software-only alternatives are typically limited to single gestures (e.g. cursor movement only) and lack robust click, scroll, and drag support. Gesture Mouse AI differentiates as a full-featured, hardware-free replacement for a physical mouse — zero extra hardware, full mouse-action parity.

6. Solution

Gesture Mouse AI is a software-only, AI-powered touchless mouse that lets users control their computer using only a standard built-in or USB webcam and natural hand gestures:

- Move cursor — point with index finger.
- Left click — pinch thumb and index finger.
- Right click — raise three fingers (index, middle, ring).
- Double click — thumb and middle finger pinch.
- Drag & drop — pinch and hold (>0.45s), move, release.
- Scroll up/down — two fingers up (peace sign), move vertically.
- Neutral/idle — fist, pauses cursor movement to prevent accidental input.

The pipeline: Camera Frame → Hand Detection (MediaPipe) → Landmark Extraction → Gesture Classification → Mouse Command Mapping (PyAutoGUI) → OS Cursor/Click Action, with Exponential Moving Average (EMA) jitter suppression and an adjustable sensitivity calibration step for reliability.

7. Why AI

Reliable, real-time interpretation of continuous hand movement into discrete cursor and click actions requires AI-based landmark detection rather than fixed rules: hands vary in size, angle, lighting, and speed across users. Google's MediaPipe hand-landmark model provides 21 3D hand-landmark tracking in real time from an ordinary RGB feed, which is then combined with custom gesture-classification logic and EMA-based jitter filtering to distinguish intentional gestures from micro-tremors and natural hand movement — something static geometric rules alone cannot do robustly across diverse users and environments.

8. Tech Stack & Plan

Stack: Python, OpenCV (video capture/processing), Google MediaPipe (21-point hand-landmark tracking), custom gesture-recognition logic (landmark position/distance classification), PyAutoGUI (OS-level cursor/click/scroll/drag mapping).

Implementation plan:

- Phase 1 (MVP) — core hand-tracking and cursor-movement pipeline.
- Phase 2 — click, right-click, double-click gesture classification.
- Phase 3 — scroll and drag-and-drop gestures.
- Phase 4 — jitter-smoothing (EMA) and sensitivity calibration.
- Phase 5 — packaging as a lightweight installable desktop app (Windows/macOS).

- Phase 6 — pilot testing with target users for feedback-driven refinement.

Milestones: stable low-latency (near real-time) tracking, a reliable gesture-classification accuracy target, and a calibration flow non-technical users can complete unassisted. Long-term roadmap includes multi-monitor support, custom gesture mapping, and enterprise features for kiosk/classroom deployment, funded initially by the founding team and hackathon/innovation-grant funding (e.g. HackX), with a premium enterprise tier planned as the future revenue source.